Mohammad Hussain

Istanbul, Türkiye · mohussain25@ku.edu.tr · +90 552 839 3308
mhdoh.github.io · github.com/MHDoh · ORCID 0009-0004-1547-3027 · linkedin.com/in/mohammad-hussain1

Research Interests

Machine learning for wireless networks: safe and risk-aware reinforcement learning, multi-timescale O-RAN control, uncertainty quantification for resource allocation, and resource-efficient models for the physical layer.

Education

Koç University, Istanbul Sep 2025 – Jul 2027 (expected)
M.Sc. in Electrical and Electronics Engineering (thesis); advisor: Prof. Sinem Coleri *GPA 3.57/4.00*
Coursework: Wireless Communications, Communication Networks, Optimization, Reinforcement Learning (ongoing)

Bilkent University, Ankara Sep 2021 – Jun 2025
B.Sc. in Electrical and Electronics Engineering; 80% merit scholarship *CGPA 3.27/4.00*

Publications

[J1] **M. Hussain**, M. Adibag, D. Gurer, G. Kalem, K. Serin, and S. Coleri, "Resource-Efficient CSI Prediction: A Gated Fusion and Factorized Projection Approach," *IEEE Communications Letters*, vol. 30, pp. 1979–1983, 2026. [DOI] [arXiv] [code]

[M1] **M. Hussain**, D. Gurer, G. Kalem, and S. Coleri, "Risk-Aware Safe Hierarchical Learning for Robust eMBB–URLLC Resource Allocation in O-RAN," manuscript in preparation for *IEEE Transactions on Wireless Communications*.

Research Experience

Wireless Networks Laboratory, Koç University Istanbul, Türkiye
Graduate Research Assistant, TÜBİTAK 1711 AI-P5G project with Turkcell and Opticoms *Oct 2025 – Present*

- **Safe hierarchical RL for O-RAN [M1].** Three PPO agents on two O-RAN timescales set slice PRB/power budgets and schedule URLLC and eMBB users; an execution-time safety layer built on conformal demand bounds and conservative channel-rate bounds corrects allocations before they are applied. Versus fixed schedulers: 61% fewer URLLC deadline misses; worst-user eMBB rate raised from 1.95 to 2.49 Mbps.
- **Lightweight CSI prediction [J1].** GRU encoder with Luong attention, gated fusion, and a factorized output head; 26% fewer parameters and $2.3\times$ the throughput of LinFormer at -13.84 dB NMSE on 3GPP TR 38.901 channels.
- **Applied private-5G control (factory site, Ankara).** Developed a controller package for the project's commercial LiteOn FlexFi gNB that lowers configured transmit power while protecting device connectivity and cell service: a data pipeline over DU statistics, PM counters, and walk-test logs; a guarded neural power selector with NETCONF actuation; and a digital twin that replays recorded walks to compare policies with fixed and rule-based baselines. Evaluated offline and prepared for factory integration.

Industry Experience

Radiologics Remote
Medical Imaging and AI Intern *Jul 2025 – Oct 2025*

- Built CT segmentation pipelines with nnU-Net v2 for 44 anatomical structures, from DICOM conversion to geometry-preserving export, and L3-centered skeletal-muscle segmentation workflows.
- Trained YOLOX detectors on 50K pediatric radiographs (recall +5–10 points); built a QCT bone-density pipeline with HU-to-BMD calibration validated by Deming regression and Bland–Altman analysis.

Modern Technology Laboratories Jeddah, Saudi Arabia
ML / Embedded Systems Intern *Aug 2024 – Sep 2024*

- Deployed a lightweight CNN for crack detection on an ESP32; built PyTorch models for concrete-strength prediction.

Selected Project

Edge vision and sensor-monitoring platform github.com/MHDoh/FYP
B.Sc. senior design project, Bilkent University, in collaboration with KAREL

- Five task-specific vision models on NVIDIA Jetson with a PyQt5 operator interface; UART ingestion of CO_2, temperature, humidity, pressure, and acceleration streams with anomaly detection; Flask/Socket.IO live dashboard.

Skills

Programming: Python, C++, MATLAB, SQL, Bash, Git, Linux, Docker, LaTeX
ML: PyTorch, TensorFlow; PPO, safe/hierarchical RL, conformal prediction, CVaR; GRU/LSTM, Transformers
Wireless: O-RAN (non-RT/near-RT RIC), 5G SA, NETCONF, QuaDRiGa, 3GPP TR 38.901 channel models
Edge and imaging: NVIDIA Jetson, ESP32, TensorFlow Lite, OpenCV, YOLOX, nnU-Net, SimpleITK
Languages: English, TOEFL iBT 114/120 (R30, L30, S26, W28; May 2025)